Unit -2 Exception Handling and Threading

Threading :-

Thread:- A small part of our program code are known as thread.

It is lightweight as compared to process

Uses of Threading :-

some common uses of threading in Python as follows.

1)Concurrent Execution: Threading enables you to execute multiple tasks concurrently. For example, you can use threads to handle network requests, perform time-consuming I/O operations, or carry out computations in parallel, making your program more efficient.

2)GUI Applications: Threading is often used in graphical user interface (GUI) applications to keep the user interface responsive while performing other tasks in the background. By running time-consuming operations in separate threads, you can prevent the GUI from freezing or becoming unresponsive.

3)Network Operations: Threading is useful for handling network-related tasks such as making HTTP requests, downloading files, or running a server. By using threads, you can initiate multiple network operations simultaneously and process their results concurrently.

4)Parallel Processing: Threading allows you to perform parallel processing, especially on multi-core or multi-processor systems. You can split a large computation-intensive task into smaller parts and execute them concurrently in separate threads to take advantage of the available computing resources.

5)Asynchronous Operations: Threading is often used in conjunction with asynchronous programming frameworks such as concurrent futures to perform non-blocking I/O operations. This allows you to initiate multiple I/O operations concurrently and continue executing other tasks while waiting for their completion.

6)Producer-Consumer Situations: Threading is commonly used in situations where multiple threads need to communicate or exchange data. For instance, in producer-consumer patterns, one thread produces data while another thread consumes it. Threading provides synchronization mechanisms such as locks, semaphores, or queues to ensure proper communication and coordination between threads.

7)Background Tasks: Threading can be used to perform background tasks while the main program continues its execution. This is useful for scenarios where you need to perform periodic tasks, maintenance operations, or data processing tasks in the background without blocking the main execution flow.

steps for creating simple thread in python :-

1)Import the required module:

Syntax :- import threading

2)**Define a function** that represents the task you want to perform in a separate thread. This function will be executed concurrently:

Syntax :- def your_task(): # Code for the task you want to perform

3)**Create a thread object**, passing the function as the target and any arguments as needed:

Syntax :- thread = threading.Thread(target=your_task)

4) **Start the thread** by calling the start() method:

Syntax :- thread.start()

5)**If needed**, you can **join the thread** to wait for its completion using the join() method:

Syntax thread.join()

Example:-1)

```
import threading
def task():
   print("Execute task")
thread=threading.Thread(target=task)
thread.start()
thread.join()
print("main thread continues executing")
```

In this example, the your_task() function represents the task you want to perform in a separate thread. The threading.Thread() class is used to create a thread object, passing your_task as the target function. The start() method starts the thread, and the join() method is called to wait for the thread's completion. Finally, the main thread continues executing after the join.

Ex.2) Creating Thread by using **thread class** with implement **run method** only

```
import time
import threading
class A(threading.Thread):
      def run(self):
           print("thread1")
class B(threading.Thread):
       def run(self):
            print("thread2")
obj1=A()
obj2=B()
obj1.start()
time.sleep(2)
obj2.start()
```

Methods of thread class :-

1)start() :-start a thread by calling start method

Ex.

```
 import threading
 def show(a,b):
       c=a+b
       print(c)
t=threading.Thread(target=show,args=(10,20))
t.start()
```

2)Run() :- Entry point for a thread.

Ex.

```
import threading
def show(a):
    print("value of a in thread=",a)
t=threading.Thread(target=show,args=(10,))
t.run()
```

3) Sleep() :-suspend thread for a specified time.

Ex.1)

```
import time
import threading
def show(a):
    print('value of an in thread1=',a)
def display(a) :
    print('value of an in thread2=',a)
t1=threading.Thread(target=show,args=(10,))
t2=threading.Thread(target=display,args=(20,))
t1.run()
time.sleep(5)
t2.run()
```

Ex. 2)

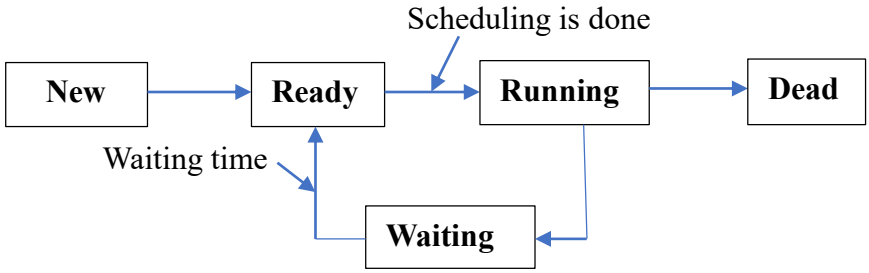
```
import threading
def show(a,b):
    print("value of a in thread=",a)
    print("value of b in thread=",b)
x=threading.Thread(target=show,args=(10,20))
y=threading.Thread(target=show,args=(30,40))
z=threading.Thread(target=show,args=(50,60))
x.run()
y.run()
z.run()
```

Thread life cycle:-

From **thread creation to thread termination** the thread goes different states are known as thread life cycle.

The life cycle has following step :-

1) New
2) Ready
3) Running
4)Waiting
5) Dead



1) New :- When we create the thread that is we create the object of that class that time it is in new state.

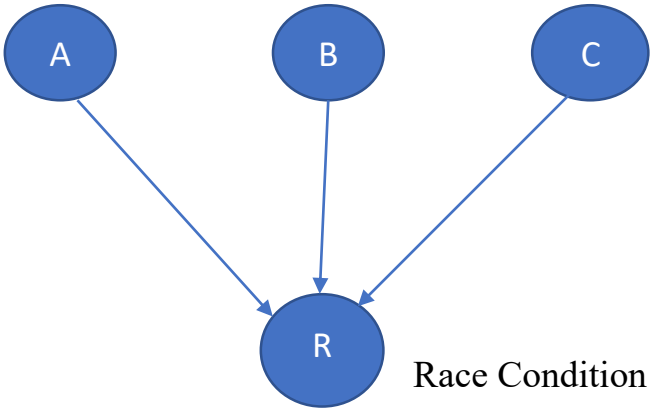
2) Ready :- After creation of thread we call the start method on thread then it is in ready state.

3) Running :-After scheduling is done it is sent to running state and in running state the actual execution is started.

4)Waiting :- During the execution of thread if any interruption is occurred i.e. if we call the sleep or join method on thread then it is waiting state.

5) Dead :-After successfully execution of thread is terminated i.e. the natural death of thread it is in dead state.

Thread Syncronization :-



Race Condition Occurs

In this mechanism the more than one thread has allow to access the common data or common resources simultaneously, at that time there will be chance of race condition. The race condition we need to implement thread synchronization.

Synchronization :-it is a mechanism which allow only one thread thread at a time access the command data or common resource i.e. execute the critical section.

In python for the thread synchronization they have provided lock class, the lock class provides following two methods.

1)**Acquire** -trade must requires the lock by using acquire method before executing it's critical section i.e. accessing the common data.

2)**Release**-the thread must need to release the lock by using release method after execution of its a critical section.

Ex.

1)

```
import threading
a=100
b=50
lock=threading. Lock()
def show():
    lock.acquire()
    c=a+b
    print("Addition of thread1=",c)
```

```
    lock.release()
def display():
    lock.acquire()
    c=a-b
    print("subtraction of thread2=",b)
    lock.release()
t1=threading.Thread(target=show)
t2=threading.Thread(target=display)
t1.start()
t2.start()
```

Exception Handling :

Before understand exception handling we have to understand about exception.

Exception: An exception is a **condition which occurs when program encounters an abnormal error.**When these exceptions occur, the Python interpreter stops the current process and passes it to the calling process until it is handled.

If these Exceptions not handled, the program will crash or stops immediately.Exceptions are of two types,

a) Built-in Exceptions b) User defined Exception

a) Built-in Exceptions: The exceptions which are already defined by python are called as built-inexceptions.

Following table shows built-in exceptions

Sr.No.	Exception Name & Description
1	Exception
	Base class for all exceptions
2	StopIteration
	Raised when the next() method of an iterator does not point to any object.
3	SystemExit

	Raised by the sys.exit() function.
4	StandardError
	Base class for all built-in exceptions except StopIteration and SystemExit.
5	ArithmeticError
	Base class for all errors that occur for numeric calculation.
6	OverflowError
	Raised when a calculation exceeds maximum limit for a numeric type.
7	FloatingPointError
	Raised when a floating point calculation fails.
8	ZeroDivisionError
	Raised when division or modulo by zero takes place for all numeric types.
9	AssertionError
	Raised in case of failure of the Assert statement.
10	AttributeError
	Raised in case of failure of attribute reference or assignment.
11	EOFError
	Raised when there is no input from either the raw_input() or input() function and the end of file is reached.
12	ImportError
	Raised when an import statement fails.
13	KeyboardInterrupt
	Raised when the user interrupts program execution, usually by pressing Ctrl+c.

14	LookupError
	Base class for all lookup errors.
15	IndexError
	Raised when an index is not found in a sequence.
16	KeyError
	Raised when the specified key is not found in the dictionary.
17	NameError
	Raised when an identifier is not found in the local or global namespace.
18	UnboundLocalError
	Raised when trying to access a local variable in a function or method but no value has been assigned to it.
19	EnvironmentError
	Base class for all exceptions that occur outside the Python environment.
20	IOError
	Raised when an input/ output operation fails, such as the print statement or the open() function when trying to open a file that does not exist.
21	IOError
	Raised for operating system-related errors.
22	SyntaxError
	Raised when there is an error in Python syntax.
23	IndentationError
	Raised when indentation is not specified properly.
24	SystemError
	Raised when the interpreter finds an internal problem, but when this error is encountered the Python interpreter does not exit.
25	SystemExit

	Raised when Python interpreter is quit by using the sys.exit() function. If not handled in the code, causes the interpreter to exit.
26	TypeError
	Raised when an operation or function is attempted that is invalid for the specified data type.
27	ValueError
	Raised when the built-in function for a data type has the valid type of arguments, but the arguments have invalid values specified.
28	RuntimeError
	Raised when a generated error does not fall into any category.
29	NotImplementedError
	Raised when an abstract method that needs to be implemented in an inherited class is not actually implemented.

b) User defined Exception : The exception defined by user is called as user defined exception.

Syntax to create user defined exception

```
raise exception
```

where raise- is a keyword which

raises exceptionexception-

any user defined exception

Exception Handling : Exception handling is a **technique to handle exception which maintains normal flow of a program .**

For exception handling following blocks are used

1) try block/clause :

In Python, exceptions can be handled using a try statement.

The critical operation which can raise an exception is placed inside the try block.The try block diverts flow of program towards next block called **except block.**

Note: In single block **only one** try block will be used.

Syntax of try block

```
   try:
--------
--------
---Logic---
```

2) except block/clause:

The except block let us handle the exception by which our program don't get terminated .For one try block we can use single or multiple except blocks.

Syntax of except block

```
except exceptiontype:
        body of except block
```

In above syntax **exceptiontype** is optional

3) finally block/clause :

The finally block lets us execute code, regardless of the result of the try- and except blocks.Finally block is an optional block.

Syntax of finally block

```
finally:
        body of finally block
```

A simple program of exception handling

```
import sys
try:
    a=10
    b=0
    c=a/b
    print("Division is",c)
except:
    print("Can not divide by zero")
finally:
    print("finally block executed")
```

Output:

```
Can not divide by zero
finally block executed
```

In above program the code is enclosed in **try** block which raise built-in exception called ZeroDivisionError because we cannot divide by zero(a/b where a=10 , b=0).The raised exception is diverted towards **except** block which will execute.After that **finally** block gets executed.

Another example of exception handling (IndexError)

```
import sys
try:
    a=[10,20,30]
    print("Division is",a[5])
except :
    print("Invalid index number")
```

Output:

```
Invalid index number
```

In above program our code kept in try block from which we are trying to access index number a[5]Which is not available.
So the built-in exception called IndexError is raised and passed towards except block where theexception is handled

User defined Exception and User defined Exception Handling / Custom Exception Handling We already seen about user defined exception and how to raise an user defined exception fromtry block.

Now we are going to handle user defined exception in

```
class myexception(Exception):
    pass
def show(age):
    try:
        if age<18:
            raise myexception
        else:
            print("yes")
    except:
            print("not")
show(10)
```

Python File Operation

File: File is a place on secondary storage where data can be stored permanently.

Since Random Access Memory (RAM) is volatile (which loses its data when the computer is turnedoff), we use files for future use of the data by permanently storing them.
In python Files are of two types ,

1) Text Files : Text file stores data in the form of Characters.

2) Binary files : Binary files stores data in the form of bytes i.e. group of 8 bits each

When we want to read from or write to a file(Both files Text and Binary), we need to open it first.When we are done, it needs to be closed .
Hence, in Python, a file operation takes place in the following order:
1) Open a file
2) Read or write (perform operation)

3) Close the file

1) Opening Files in Python :-

Python has a built-in open() function to open a file. This function returns a file object it is used toread, write or modify the file .

The open() function takes two parameters; *filename*, and file opening *mode*.

```
f = open("test.txt")    # open file in current directory
f = open("C:/Python38/README.txt")  # specifying full path
f = open("test.txt",'w')  # write in text mode
```

We can specify the mode while opening a file. In mode, we specify whether we want to read **r**,write **w** or append **a** to the file. We can also specify if we want to open the file in text mode orbinary mode.

The default mode is reading in text mode. In this mode, we get strings when reading from the file.On the other hand, binary mode returns bytes and this is the mode to be used when dealing with non-text files like images or executable files.

File Opening Modes :-

Mode	Description
r	Opens a file for reading. (default)
w	Opens a file for writing. Creates a new file if it does not exist or truncates the file if it exists.
x	Opens a file for exclusive creation. If the file already exists, the operation fails.
a	Opens a file for appending at the end of the file without truncating it. Creates a new file if it does not exist.
t	Opens in text mode. (default)
b	Opens in binary mode.
+	Opens a file for updating (reading and writing)

2) **Read or write (perform operation) :** We will reading or writing text and binary files but beforethat we are going to understand closing operation

3) **Close the file :** After reading or writing a file we need to close the file as ,

```
f.close()
```

Python File Methods :

There are various methods available with the file object. Here is the complete list of methods intext mode with a brief description:

Method	Description
close()	Closes an opened file. It has no effect if the fileis already closed.
detach()	Separates the underlying binary buffer fromthe TextIOBase and returns it.
fileno()	Returns an integer number (file descriptor) ofthe file.
flush()	Flushes the write buffer of the file stream.
isatty()	Returns True if the file stream is interactive.
read(n)	Reads at most n characters from the file. Readstill end of file if it is negative or None.
readable()	Returns True if the file stream can be readfrom.
readline(n=-1)	Reads and returns one line from the file. Readsin at most n bytes if specified.
readlines(n=-1)	Reads and returns a list of lines from the file. Reads in at most n bytes/characters if specified.
seek(offset,from=SEEK_SET)	Changes the file position to offset bytes, inreference to from (start, current, end).
seekable()	Returns True if the file stream supports randomaccess.
tell()	Returns the current file location.
truncate(size=None)	Resizes the file stream to size bytes. If size isnot specified, resizes to current location.
writable()	Returns True if the file stream can be writtento.
write(s)	Writes the string s to the file and returns thenumber of characters written.
writelines(lines)	Writes a list of lines to the file.

Now let's see Writing and reading operation on **Text** file first,

Text File Writing :

In order to write into a file in Python, we need to open it in write **w**, append **a** or exclusivecreation **x** mode.

We need to be careful with the w mode, as it will overwrite into the file if it already

exists. Due tothis, all the previous data are erased.

Program to write data in Text file(w)

```
f=open('test.txt','w')
f.write("Hi")
f.close()
```

-In above program the file is opened by open() method. The open method contains two argumentsfirst one is 'test.txt'(it us a file name which is stored in its default location i.e.python folder present in C drive) and second argument is 'w' file opening mode as write mode.

-After that we write a string **Hi** by using write() method.And after that we closed our file by close() method.

Text File Reading(r) : To read a file in Python, we must open the file in reading **r** mode.There are various methods available for this purpose.

We can use the read(size) method to read in the size number of data.

If the size parameter is not specified, it reads and returns up to the end of the file.

```
f = open("Test.txt", "r")
m=f.read()
f.close()
print(m)
```

Reading Multiline Text File Line by Line:

we can read multiline text file line by Line forthat readlines() method is used.

```
f = open("Test.txt", "r")
m=f.readlines()
for x in m:
        print(x)
f.close()
```

Text File Appending(a/a+)

In order to append a new line to the existing file, open the file in append mode, by usingeither 'a' or 'a+' as the access mode. The definition of these access modes are as follows:

-**Append Only ('a'):** Open the file for writing. The file is created if it does not exist. The handle ispositioned at the end of the file. The data being written will be inserted at the

end, after the existing data.

-**Append and Read ('a+'):** Open the file for reading and writing. The file is created if it does notexist. The handle is positioned at the end of the file. The data being written will be inserted atthe end, after the existing data.

When the file is opened in append mode, the handle is positioned at the end of the file. The databeing written will be inserted at the end, after the existing data. Let's see the below example to clarify the difference between write mode and append mode.

```
f=open('test.txt','a')
f.write("How are You?")
f.close()
```

Output :Hi How are you

In above program we opened our previous file test.txt (which contain a string 'Hi') in appendmode.
After that the string How are you is append on file using write() function.The write function appends How are you at the end of string Hi .

Binary File writing:

Before writing binary file first we have to open a file by same way but the file opening modeshould be wb .
After that we can write a binary file by write() or writelines() method.
Program to write Binary file

```
f=open("binfile.bin","wb")
num=[5, 10, 15, 20, 25]
arr=bytearray(num)  #converting to bytearray type
f.write(arr)
f.close()
```

Binary File reading:

Before reading binary file first we have to open a file with file opening mode **rb** .After that we can read a binary file by read() or readlines() method.

```
f=open("binfile.bin","rb")
p=f.read()                          #reading binary file
num=list(p)              # conversion of binary data to list type
print (num)
f.close()
```

seek() and tell() methods:

seek() method :

In Python, seek() function/method is used to **change the position of the File Handle** to a given specific position. File handle is like a cursor, which defines from where the data has to be read orwritten in the file.

Syntax:

```
f.seek(offset, from_what)
```

where f is file pointer

Parameters:

Offset: Number of postions to move forward from_what: It defines point of reference.

The reference point is selected by the from_what argument.

It accepts three values:

0: sets the reference point at the beginning of the file

1: sets the reference point at the current file position

2: sets the reference point at the end of the file

By default from_what argument is set to 0.

tell() method :

The tell() method returns the file handlers current position in a file stream.

Syntax

```
f.tell()
```

where f is file pointer

Example of seek() and tell() method

Note: test.txt file is already present in secondary storage in that file Sangola(data) is present.

```
f=open("F:/test.txt",'r')
f.seek(3)
v=f.tell()
print('file pointer is on',v)
str=f.read()
print(str)
f.close()
```

Output: file pointer is on 3

gola

Program of file with exception handling

```
try:
   f=open('test.txt','r')
   s=f.read()
   f.close()
   print(s)

except:
   print('can not perform operation on file')
```

Python Directory and Files Management :

If there are a large number of files to handle in our Python program, we can arrange our code within different directories to make things more manageable.

A directory or folder is a collection of files and subdirectories. Python has the os module thatprovides us with many useful methods to work with directories (and files as well).

Get Current Directory :-

We can get the present working directory using the getcwd() method of the os module. This method returns the current working directory in the form of a string. We can also usethe getcwdb() method to get it as bytes object.

```
import os

os.getcwd()
os.getcwdb()
```

Making a New Directory :-

We can make a new directory using the mkdir() method.

This method takes in the path of the new directory. If the full path is not specified, the newdirectory is created in the current working directory.

```
>>> os.mkdir('test')

>>> os.listdir()
['test']
```

Renaming a Directory or a File :-

The rename() method can rename a directory or a file.

For renaming any directory or file, the rename() method takes in two basic arguments: the oldname as the first argument and the new name as the second argument.

```
>>> os.listdir()
['test']

>>> os.rename('test','new_one')

>>> os.listdir()
['new_one']
```

Removing Directory or File :-

A file can be removed (deleted) using the remove() method. Similarly, the rmdir() method removesan empty directory.

```
>>> os.listdir()
['new_one', 'old.txt']

>>> os.remove('old.txt')
>>> os.listdir()
['new_one']

>>> os.rmdir('new_one')
>>> os.listdir()
[]
```

Note: The rmdir() method can only remove empty directories.

In order to remove a non-empty directory, we can use the rmtree() method inside the shutil module.

```
>>> os.listdir()['test']

>>> os.rmdir('test')
Traceback (most recent call last):
...
OSError: [WinError 145] The directory is not empty:'test'
>>> import shutil

>>> shutil.rmtree('test')
>>> os.listdir()[]
```